



A composite polyphenol-rich extract improved growth performance, ruminal fermentation and immunity, while decreasing methanogenesis and excretion of nitrogen and phosphorus in growing buffaloes

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Abstract

The effects of a composite polyphenolic-rich extract (CPRE) on ruminal fermentation, nutrient utilisation, growth performance, excretion of nitrogen and phosphorus and methane emission were studied in growing buffaloes. Four herbal dry extracts prepared from *Acacia arabica* (babul; bark), *Acacia catechu* (cutch; bark), *Punica granatum* (pomegranate; peel) and *Eugenia jambolana* (Indian blackberry; seeds) were mixed in an equal proportion (1:1:1:1) to prepare the CPRE that contained mainly phenolic compounds (146 g/kg), flavonoids (41.7 g/kg) and saponins (40.5 g/kg). First, in vitro tests were performed for ruminal fermentation and feed degradability using ruminal fluid as inocula and CPRE at 0 to 40 g/kg substrate to decide an optimal dose of CPRE for an in vivo study on buffaloes. In the animal study, 20 buffaloes were randomly assigned to two groups ($n = 10$)—a control diet and a CPRE diet (control diet added with extra 20 g/kg of CPRE). The in vitro tests suggested that addition of CPRE at 20 g/kg substrate increased degradability of substrate, short-chain fatty acid concentration and propionate proportion, and reduced methane production, acetate proportion, acetate:propionate ratio and ammonia concentration in fermentation media, which were also noted in the rumen of buffaloes. Feeding CPRE to buffaloes did not affect feed intake, but increased daily body weight gain, dry matter and crude protein digestibility and nitrogen and phosphorus retention in the body. Total bacteria, methanogens and protozoal numbers were similar between two groups, but *Fibrobacter succinogenes* increased in the rumen of buffaloes fed CPRE. Concentrations of total, essential, non-essential and glucogenic amino acids were greater in the plasma of CPRE-fed buffaloes. Cell-mediated immune response improved in the CPRE-fed buffaloes compared with the control group. Estimated methane production and excretion of nitrogen and phosphorus per unit of body weight gain decreased in the CPRE group. The comprehensive results of this study clearly suggested that the composite polyphenol-rich feed additive at 20 g/kg diet improved growth performance, ruminal fermentation, immunity and plasma amino acids profile, whereas it reduced indicators of environmental impacts of buffalo production.

Keywords Polyphenols · Buffalo · Ruminal fermentation · Growth performance · Immunity · Environmental indicator

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Introduction

Livestock production supports the livelihoods and nutritional food security of the majority of human populations, particularly, in the middle- and low-income countries of the world (Herrero et al. 2013). However, global livestock biomass is double the human biomass (Bar-On et al. 2018), and utilises 58% of human-food biomass globally (Krausmann et al. 2008). Concurrently, livestock production is considered for many negative consequences including climatic change, pollution, poor animal welfare, biodiversity loss, land-use change, human illness owing to zoonosis, ill health because

of high consumption of livestock-derived foods and antibiotic resistance of pathogenic microorganisms due to use of antibiotic feed additives (Chowdhury et al. 2018; Mueller and Lassaletta 2020; Salmon et al. 2020). Livestock supply chains contribute 14.5% of total greenhouse gas emissions from anthropogenic activities (Gerber et al. 2013), one-third of reactive nitrogen emission (Mueller and Lassaletta 2020) and phosphorus pollution (Guo et al. 2019; Rothwell et al. 2020). Livestock production in 2050 may elevate human food biomass use by 21%, direct greenhouse gas emissions by 39%, and reactive nitrogen emission by 36% beyond the 2000 year levels (Pelletier and Tyedmers 2010).

With the contentious issue of ruminant production between nutritional security and environmental damage, animal scientists look for several solutions of the negative consequences, notably the use of natural plant bioactives instead of antibiotics, mitigation of enteric methane production (Patra et al. 2017) and reduction of nitrogen and phosphorus excretion (Abbasi et al. 2018; Grosse Brinkhaus et al. 2016; Guo et al. 2019; Patra et al. 2020). In this context, many phytochemicals were evaluated with contrasting outcomes on animal performance, nutrient utilisation, methane inhibition and nitrogen excretion (Grosse Brinkhaus et al. 2016; Makkar et al. 2007; Patra and Saxena 2010). Phenolic compounds are one of phytochemical groups that are abundantly present in many plants and confer several biological and pharmaceutical benefits (Patra 2012; Wang et al. 2020). They have been exploited in some experiments to reduce methane emission and protein utilisation (Ineichen et al. 2019; Sinz et al. 2019).

Waste biomass from medicinal plant materials as feed additive shows promise to modulate ruminal fermentation, health, ruminant performance and methane mitigation (Singh et al. 2021; Singla et al. 2021). Extraction of functional bioactives from waste products and their utilisation as feed additives to overcome negative consequences on ruminant production would provide double benefits—waste valorisation and environmental concern reduction. In this context, the bark of *Acacia arabica* (babul tree) and *Acacia catechu* (cutch tree), peel of *Punica granatum* (pomegranate fruit) and seeds of *Eugenia jambolana* (Indian blackberry tree) are the waste products of wood and fruit industries, but they contain numerous functional bioactive principles including rich sources of polyphenolic compounds (Baliga et al. 2011; Kumar et al. 2019; Mohammad et al. 2014; Singh et al. 2018). These plant materials are integral parts in various ethnomedicines (Baliga et al. 2011; Mohammad et al. 2014; Rajvaidhya et al. 2012; Singh et al. 2018). Due to rich concentrations of polyphenolic compounds with especially antimicrobial, antioxidant and immune-stimulating actions, extracts from these plant materials could exert beneficial effects on production and health of buffaloes. Moreover, a composite mixture of these plant extracts could broaden

the spectrum of polyphenolic compounds including some other phytochemicals providing better multi-functionality in responses (Patra and Yu 2015; Sinz et al. 2019). There are limited studies to investigate comprehensively the production performance, methane emission and excretion of nitrogen and phosphorus in ruminants using a composite polyphenol-rich extract (CPRE) from waste biomass of medicinal plants. Therefore, a comprehensive study was carried out to investigate the effects of a CPRE on ruminal fermentation, growth performance, nutrient utilisation, immunity, plasma amino acid profile, methane emission and excretion of nitrogen and phosphorus in growing buffaloes.

Methods

Composite polyphenol-rich feed additive

The four different types of dry plant extracts that were prepared from bark of *Acacia arabica* (babul), bark of *Acacia catechu* (cutch), peel of *Punica granatum* (pomegranate) and seeds of *Eugenia jambolana* (jambol or Indian blackberry) were purchased from Konark Herbs and Health Care, Mumbai, India. The dry plant extracts were mixed in an equal proportion (1:1:1:1) to prepare a composite polyphenol-rich feed additive to broaden the spectrum of bioactive principles. The concentrations of total phenolic compounds, flavonoids, total condensed tannins, saponins and ascorbic acid were determined in the individual extracts and CPRE following the methods described earlier (Singla et al. 2021).

Experimental diet

A control total mixed ration (TMR) consisting of green berseem (*Trifolium alexandrinum*), wheat straw and a concentrate mixture was prepared by mixing in the proportion of 36, 24 and 40% on dry matter (DM) basis (Table 1). The roughage to concentrate proportion was maintained at 60:40. This TMR was used as a substrate for in vitro experiments and for feeding of buffaloes to evaluate CPRE (in vivo experiment).

In vitro fermentation characteristics, gas kinetics and methane production

The fermentation kinetics, gas kinetics and methane production from TMR with different levels of CPRE were evaluated by a gas production method using graduated glass syringes (Haberle Labortechnik, Germany) (Menke and Steingass 1988). Dietary CPRE doses were 10, 20, 30 and 40 g/kg dry matter (DM) including a control (without CPRE). The respective CPRE treatments were prepared by adding CPRE as top-dressing to the control TMR.

Table 1 Ingredient composition of control total mixed ration (TMR) and concentrate mixture and chemical composition of control TMR fed to buffalo calves

Ingredient composition	g/kg DM	Chemical composition of TMR	g/kg DM
Control TMR composition ^a		Total ash	93.9
Berseem green	360	Organic matter	906
Wheat straw	240	Crude protein	141
Concentrate mixture	400	Ether extract	26.0
Concentrate mixture		Neutral detergent fiber	513
Maize grains	250	Acid detergent fiber	306
Wheat grain	150	Acid detergent lignin	51.4
Mustard cake (expeller)	100	Cellulose	255
Deoiled mustard cake	50	Hemicellulose	207
Soybean meal	100	Gross energy, MJ/kg	20.1
Deoiled rice bran	210	Phosphorus	3.80
Rice bran	100	NDICP	48
Urea	3	ADICP	36
Mineral mixture ^b	20	Metabolizable energy (MJ/kg) ^c	8.95
Salt (sodium chloride)	10	tdCP (g/kg) ^c	104
Sodium bicarbonate	7		

DM dry matter, NDICP neutral detergent insoluble crude protein, ADICP acid detergent insoluble crude protein, tdCP truly digestible crude protein

^aControl TMR was added with composite polyphenol-rich extract at 20 g/kg as top-dressing in the experimental group

^bMinerals content (g/kg mineral mixture on DM basis): calcium, 200; phosphorus, 120; magnesium, 50; sulphur, 20; iron, 4; copper, 1; zinc, 8; manganese, 1.2; cobalt, 0.12; iodine, 0.26

^cCalculated values as per NRC (2001)

The source of microbial inocula was ruminal fluid collected from three rumen cannulated Murrah buffalo bulls (age 7–8 years; body weight 558 ± 6.08 kg) which were fed the TMR (control diet without CPRE) as per standard nutritional requirements (ICAR 2013). The ruminal fluid was collected pre-prandial in a pre-warmed double walled (thermo) flask (flushing with carbon dioxide gas) and blended for 3 min in a blender at 39 °C. The ruminal fluid after straining through two-layered muslin cloth was mixed in a ratio of 1:2 with buffer solution (Menke and Steingass 1988) to prepare the incubation medium under continuous flow of carbon dioxide at 39 °C. The TMR of about 375 mg was weighed in syringes with different levels of CPRE and incubated in a water bath maintained at 39 °C after dispensing 30 mL of incubation medium for 24 h (swirled hourly). Three syringes were prepared for each of the treatments and blanks (without any substrate). The incubation was ended after a stipulated period, and gas volume in syringes was noted and net gas production (NGP) was calculated after blank correction. In vitro true organic matter degradability (IVTOMD) and neutral detergent fiber degradability (IVNDFD) were determined after blank correction (Blummel et al. 1997). The microbial biomass production (MBM, mg/g truly OM degraded substrate) was calculated subtracting net gas volume (mL/g substrate) $\times 2.2$ from in vitro true degraded OM (mg/g substrate), where 2.2 is the stoichiometric factor (Blummel

et al. 1997). The partition factor (PF) was also calculated as described by Blummel et al. (1997).

To determine gas production kinetic parameters, the syringes were incubated with 200 mg substrate (containing different levels of CPRE) and 30 mL fermentation medium (Menke and Steingass 1988) in a water bath (39 °C) for 72 h. The gas volume was recorded at 2-h intervals up to 12 h and then 12-h intervals up to 72 h after incubation. The gas volume data after blank correction were fitted to a nonlinear model of SAS package (2001) to find out gas production kinetics parameters as described by Pal et al. (2015). The model used was as follows:

$$Y = Y_{\max} \times [1 - \exp^{-K(T-L)}]$$

where Y is gas volume (mL/g DM) at time T (hour), $Y_{\max}$ is maximum potential of gas production (mL/g DM), K is fractional gas production rate per hour and L is lag time (h). The half time of the greatest gas volume ($T_{1/2}$) was calculated as $\ln(2)/K$ (Singh et al. 2021).

The incubations were run again with the same way as for the gas production kinetics and were terminated at the $T_{1/2}$ of each treatment. The gas production in glass syringes (Haberle Labortechnik, Germany) was recorded. A gas sample of 100 μ L from the gas phase was injected into a gas chromatograph for measurement of methane concentration using standard calibration gas composed of 50% methane

and 50% carbon dioxide (Singla et al. 2021). After termination of incubation (at $T_{1/2}$), a 5-mL aliquot of fermentation fluid was processed by mixing with 1 mL of meta-phosphoric acid solution (25%) and left at room temperature for 1 h. The contents were centrifuged at 5500 rpm for 10 min and the supernatant was stored at -20°C until analysis of short-chain fatty acid (SCFA).

Feed intake and growth performance

Twenty Murrah male buffaloes with average body weight ($\pm$ standard error) of 155 ± 0.28 kg and average age of 12.4 ± 0.27 months were randomly assigned to two groups ($n=10$): without CPRE (control) or with CPRE (treatment; 20 g/kg DM intake). The experimental period consisted of a 10-day TMR adjustment period followed by a 90-day data recording period, i.e. actual growth trial. The growing animals were fed a control TMR composed of green berseem, wheat straw and a concentrate mixture in the proportions of 36%, 24% and 40% on dry matter (DM) basis similar to in vitro substrate (TMR). The ratio of roughage to concentrate was maintained at 60:40 throughout the experimental period. The proportion of wheat straw and green forage was 40:60 (Table 1). The CPRE at 20 g/kg as top-dressing was mixed thoroughly with the concentrate mixture before inclusion in the TMR. All buffaloes were kept in a well-ventilated shed under similar management conditions and stall fed individually at 9.30 AM daily as per their nutrient requirements (ICAR 2013). Drinking water was provided ad libitum twice a day throughout the feeding trial. The animals were taken out in an open space for exercise (1 h) daily. Body weight of buffaloes was recorded three consecutive days at 14-day intervals to obtain average daily body weight gain (ADG) and revise their feeding schedule every 14 days.

Metabolic trial and blood collection

A 6-day metabolic trial on 16 animals (8 buffaloes/treatment) was performed after 70 days of feeding. The trial was conducted in metabolic cages where total urine and faeces were automatically collected in plastic cans and drums, respectively. The quantity of TMR offered and leftover was noted to workout dry matter intake. After an adaptation period of 2 days, total amounts of faeces and urine were collected, weighed and recorded daily. On the basis of urine output during adaptation period, about 150 mL of 20% sulphuric acid was added to each plastic can before it was placed under metabolic cage for collection of urine. This was done to reduce the pH of urine below 2.0 immediately after collection. About 10 mL/L of total volume of fresh urine was sampled from plastic cans after thoroughly mixing and stored in a deep freezer for further analysis. Similarly, a suitable aliquot of total faeces was sampled in containers

containing 20% sulphuric acid solution (25 mL) for analysis of crude protein (CP) content. Also, a suitable amount of faeces was dried at 60°C for 48 h in a hot air oven, and then pooled together for each animal separately. The dried faecal samples were stored after grinding for nutrient analysis to determine apparent nutrient digestibility and phosphorus excretion. On the last day of metabolic trial, blood samples were collected from individual animals in the heparinized tubes (4-h postprandial), centrifuged at 3500 rpm for 15 min and plasma samples were stored for further analysis. After plasma separation, the cell sediments were washed thrice with sodium chloride solution (0.9%) and were haemolysed using distilled water (ninefold) to prepare 10% hemolysate for analysis of superoxide dismutase (SOD).

Immune response

The delayed-type hypersensitivity technique was used to examine the cell-mediated immune response in male buffaloes (Abbas et al. 2014) between 78 to 80 days of experimental period. The neck regions of animals were cleaned and shaved at two different sites (approximately at the distance of 4 cm) 24 h prior to the phytohaemagglutinin-P (PHA-P; HiMedia Laboratories, Mumbai, India) antigen injection. PHA-P (150 μg dissolved 200 μL in PBS solution) was injected subcutaneously at the selected neck region. The skin thickness (swelling due to leukocyte recruitment) at the site of PHA-P injection was measured by using a vernier calliper at 12-h intervals up to 72 h.

Collection of ruminal fluid

Rumen fluid samples were withdrawn from both control as well as experimental animals using oro-ruminal collection technique for two consecutive days (4-h postprandial) before termination of the feeding trial. A Pedal Suction Apparatus (Model; 7B, Ishneel Health Care Pvt. Ltd. Mumbai, India) containing a suction pump connected with a stomach tube was used for ruminal fluid sampling through oro-ruminal passage. The ruminal fluid samples were filtered through double-layered-muslin cloth and stored at 4°C after arresting microbial activity (by mixing saturated mercuric chloride solution) for determination of different nitrogen fractions and SCFA concentration. The samples for SCFA analysis were prepared as described earlier. Equal quantity of strained ruminal fluid from both days' collections was pooled animal-wise before their analysis. To study abundances of genes, ruminal fluid (1 mL) was mixed with 1 mL of RNAiso Plus (Cat. No. 9108; TaKaRa Bio Inc., Kusatsu, Japan) immediately after collection and stored at -80°C for microbial analysis.

Analysis of samples

Chemical composition of TMR, CPRE, orts and excreta

The TMR, CPRE, orts and faeces samples were analysed for DM, total ash, ether extract following AOAC International (2007) methods. The nitrogen content in fresh faeces, urine and rumen liquor samples were determined by Kjeldahl method (method no. 976.05). Total ash content was subtracted from DM content to obtain organic matter (OM) content. The forage detergent system was used to assay fiber fractions (Van Soest et al. 1991). The components obtained after subtracting acid detergent fiber (ADF) content from neutral detergent fiber (NDF) content and acid detergent lignin (ADL) content from ADF represented hemicellulose and cellulose concentrations, respectively. The concentrations of neutral detergent insoluble crude protein and acid detergent insoluble crude protein were determined in the NDF and ADF components, respectively as per Van Soest et al. (1991), followed by Kjeldahl method (method no. 976.05). Gross energy (GE) values of TMR and CPRE were measured using a bomb calorimeter (Toshniwal Bros, New Delhi, India). The phosphorus content in TMR, faeces and urine was analysed by phosphomolybdenum blue complex colorimetric method of Koenig and Johnson (1942).

Ruminal fluid analysis, fermentation efficiency and prediction of in vivo methane production

The ruminal fluid samples from in vitro and in vivo experiments were analysed following the standard method (Cottyn and Boucque 1968). From SCFA profile, fermentation efficiency (FE), SCFA utilisation index (SCFA-UI) and methane production were calculated as per following equations (Orskov 1975, Moss et al. 2000, Williams et al. 2019):

$$\text{FE, \%} = (0.622 \times A + 1.092 \times P + 1.56 \times B) \times 100 / (A + P + 2B)$$

$$\text{SCFA - UI} = (A + 2B + V) / (P + V)$$

$$\text{Methane production, g/mol SCFA} = 0.45 \times A - 0.275 \times P + 0.40 \times B$$

$$\text{Methane production, g/kg DM intake} = 3.28 \times (A + B) / P + 7.6$$

Where A, B, P and V stand for molar proportions of acetate, butyrate, propionate and valerate, respectively. Methane energy value of 55.4 kJ/g was taken for calculation of energy loss as methane (Brouwer 1965).

In strained ruminal fluid samples, the total precipitable nitrogen (TCA-N) concentration was determined by a method described by Cline et al. (1958) and subtraction of TCA-N from total nitrogen represented non-protein nitrogen.

Ruminal ammonia concentration was measured using the micro-Kjeldahl method (AOAC International 2007).

Biochemical and enzyme profiling of blood plasma

The various biochemical indices (glucose, triglycerides, total protein, albumin, cholesterol, triglyceride, blood urea nitrogen, creatinine, alanine aminotransferase (ALT) and aminotransferase (AST) in plasma samples and superoxide dismutase (SOD) from blood hemolysate samples) were analysed on a semi-automatic biochemical analyser using Erba Mannheim kits (Transasia Bio-Medical Ltd., India) as per the manufacturer's diagnostic protocols.

Amino acid profiling of blood plasma

For the extraction of free amino acids, a 50 µL of plasma samples and 50 µL of sulfo-salisilic acid (10%) were taken in micro-centrifuge tube and vortexed for 1 min. The mixture was centrifuged at 14,000 rpm for 8 min. The supernatant was collected for amino acid analysis. The manufacturer's protocol was followed to analyse the derivatized free amino acids with AccQ•Tag™ ultra reagent (6-aminoquinolyl-N-hydroxysuccinimidyl carbamate). Briefly, 70 µL of borate buffer, 10 µL of supernatant and 20 µL of AccQ•Tag™ ultra reagent were mixed in an Eppendorf tube and vortexed at room temperature (Armenta et al. 2010). Then, this mixture was incubated for 10 min in a water bath at 55 °C and at least 2 min at room temperature before it was used for injection.

The free amino acid concentrations were assayed using liquid chromatography on Water's Acquity UPLC system (Waters India Pvt. Ltd.), equipped with an autosampler, binary solvent manager, column heater and a tunable UV (TUV) detector. The Waters AccQ•Tagultra-column (2.1 mm × 100 mm, 1.7 µm particles) was used as a separation column. The mobile phase flow rate was kept at 0.7 µL/minute, whereas the column temperature was adjusted at 55 °C. Eluent A and B were 10% AccQ•Tag ultra-concentrate solvent A and 100% AccQ•Tag ultra-concentrate solvent B, respectively. Prepared samples (1 µL) were injected into the UPLC system. The protein hydrolysate standards (Waters Corporation, Milford, U.S.A.) contained 2.5 µmole/mL for each amino acid except for cysteine (1.5 µmole/mL). Plasma concentrations of individual amino acids were determined by comparing the area of the sample with the area of diluted standards (100 µL standard: 900 µL 0.1 N HCl).

Different amino acids were added to form amino acid groups, namely essential amino acid (EAA) included Thr + Val + Leu + Met + Ile + Phe + His + Arg + Lys; the non-essential amino acid (NEAA) included Glu + Asp + Ser + Pro + Gly + Cys + Ala + Tyr, glucogenic amino acids (GAA) included Val + Met + Thr + His + Arg + Glu + Gly + Ser + Ala + Asp + Pro + Cys, ketogenic amino acid (KAA) included

Lys + Leu; and branched-chain amino acid (BCAA) included Val + Ile + Leu.

RNA isolation and real time PCR

The relative quantification of ruminal microbes, i.e. total bacteria, *Ruminococcus flavefaciens*, *Fibrobacter succinogenes*, total protozoa and total methanogens (six from each group) was carried out with real time qPCR. The total RNA in the ruminal fluid samples (2 mL) was extracted by Trizol method (cat. no. 15596018) and subsequently converted to cDNA by using TruScript First Strand cDNA Synthesis Kit (BioCat GmbH, Heidelberg, Germany) according to the manufacturer's instruction. The qPCR was performed by using a Real Time System (CFX 96, Bio-Rad iCycler Thermocycler, Bio-Rad, Hercules, CA) with fluorescence detection of SYBR green dye. The primer sequences for total bacteria, *Fibrobacter succinogenes*, *Ruminococcus flavefaciens* (Denman and McSweeney 2006), total methanogens (Denman et al. 2007), total protozoa (Sylvester et al. 2005), and bacterial 16S rRNA gene (5'-AGAGTTTGATCC TGGCTCAG-3' and 5'-GGTTACCTTGTTACGACT T-3'; universal primers) as a reference (Zhang et al. 2020) were used from the literature. The reaction was performed using aliquots of cDNA and SYBR Green qPCR Master mix (QIAGEN Science, Germantown, Maryland, USA) in duplicate. The results of real time PCR were then calculated using: $\Delta Ct = Ct [\text{target}] - Ct [\text{reference}]$ and $\Delta\Delta Ct = [\Delta Ct \text{ of experiment}] - [\Delta Ct \text{ of control}]$. The fold change in expressions of genes was calculated by using formula $2^{-\Delta\Delta Ct}$.

Statistical analysis

The data were analysed for the variables except repeated measure data using generalised linear model (SPSS 2009) in a completely randomised design with the dose of CPRE as main effect and differences in mean were tested by Tukey's b test (in vitro evaluation). The dose effect was worked out by using quadratic and linear polynomial contrasts.

The exponential model ($Y_T = Y_{\max} \times [1 - \exp^{-K(T-L)}]$) was fitted to net gas production data in the NLIN procedure of SAS (2001). Cell-mediated immune response was analysed with treatment, hour and their interaction as main effects, animal as the random effect using MIXED procedures (SAS 2001) and time as the repeated measures. Covariate structure type = arh(1) resulted in the best fit model. Because treatment and hour interaction was significant, 'slice' option was used

in the SAS program to find out the time point where treatment effect was significant.

Results

Bioactive compounds in CPRE

The feed additive used in this study had high concentration of total phenolic compounds (146 g/kg) with a high amount of flavonoids (Table 2). It also contained saponins and condensed tannins. In the in vivo study, 20 g/kg of CPRE was added in the TMR that contained phenolics at 2.92 g/kg TMR.

In vitro gas production, degradability and microbial biomass production (in vitro experiment)

Net gas production at 24 h was not influenced by different doses of CPRE (Table 3). Increasing doses of CPRE increased IVTDOM, IVNDFD and MBP linearly and quadratically ($P < 0.01$) with the greatest values at the doses of 10 to 20 g/kg diet. The PF values lowered linearly with increasing doses of feed additive.

In the gas production kinetics study, gas production was not affected in different hours except at 60 and 72 h where gas production increased in a quadratic manner with the highest gas production at 10 to 30 g/kg diet (Table 4). The gas production kinetic parameters, i.e. L , K and $T_{1/2}$, were unchanged; whereas $Y_{\max}$ showed a quadratic effect ($P = 0.003$) with the greatest value at 20 to 30 g/kg diet. Methane production expressed at mL/g incubated DM of in vitro degraded DM or in vitro true degraded OM decreased linearly ($P < 0.01$) with increasing doses of CPRE. In vitro ammonia-N concentration in the fermented media decreased in a linear and quadratic manner ($P < 0.01$) with the lowest concentration at 30 to 40 g/kg diet of CPRE.

In vitro ruminal short-chain fatty acid profile (in vitro experiment)

Concentrations of total SCFA (linear and quadratic effect; highest at 20 to 30 g/kg), acetate (linear effect), propionate (linear and quadratic effect; highest at 20 to 30 g/kg), butyrate (linear and quadratic effect; highest at 30 to 40 g/kg) increased with greater doses of feed additive (Table 5). The concentration of total branched-chain fatty acids (BCFA) was not altered. The acetate to propionate ratio decreased linearly and quadratically with lowest value at CPRE doses of 20 to 30 g/kg. The molar proportion of

Table 2 Chemical composition of composite polyphenolic-rich extract (CPRE) and bioactive components of individual and (CPRE) used in the study

Chemical composition		Bioactive components, g/kg					
CPRE ^a	g/kg, unless stated	Item	<i>Acacia arabica</i>	<i>Acacia catechu</i>	<i>Punica granatum</i>	<i>Eugenia jambolana</i>	CPRE ¹
Total ash	23.0	Total phenolics	113	158	157	157	146
Organic matter	977	Condensed tannins ^b	2.6	40.7	0.8	3.6	11.9
Crude protein	19.7	Saponins	72.6	20.8	25.9	42.9	40.5
Ether extract	4.0	Ascorbic acid	3.6	28.3	6.1	24.8	15.7
Neutral detergent fiber	23.0	Flavonoids	94.7	9.6	56.6	6.1	41.7
Acid detergent fiber	8.30						
Acid detergent lignin	0.10						
Cellulose	8.20						
Hemicellulose	14.7						
Gross energy, MJ/kg	19.5						

^aCPRE contained *Acacia Arabica* bark extract, *Acacia catechu* bark extract, *Punica granatum* peel extract and *Eugenia Jambolana* seed extract at a 1:1:1:1 ratio

^bLeucocyanidin equivalent

acetate and total BCFA lowered linearly, whereas the molar proportion of propionate and butyrate elevated linearly with increasing doses of CPRE. The estimated fermentation efficiency improved linearly and SCFA utilisation index lowered linearly due to addition of CPRE.

Growth performance, nutrient digestibility, nitrogen and phosphorus balance (in vivo experiment).

Intake, final body weight gain and feed efficiency were similar, but average daily gain increased ($P=0.035$) due to CPRE feeding compared with the control (Table 6). Apparent digestibility of DM ($P=0.039$) and CP ($P=0.021$) improved and digestibility of OM ($P=0.056$) and cellulose ($P=0.070$) tended to improve in the CPRE group. Digestibility of NDF and ADF was not affected by CPRE feeding in growing buffaloes.

Nitrogen intake, faecal nitrogen, urinary nitrogen and absorption were not influenced ($P>0.10$), but nitrogen retention ($P=0.011$) and apparent biological value of absorbed protein ($P=0.032$) was better for the CPRE group than the control group. Phosphorus intake, faecal phosphorus and urinary phosphorus excretion were similar ($P>0.10$) between two groups, but phosphorus absorption ($P=0.020$), retention ($P=0.020$) and absorption percent ($P=0.003$) increased due to CPRE feeding.

Ruminal fermentation and select microbial population (in vivo experiment)

The concentrations of total SCFA, propionate and butyrate increased ($P<0.05$), isobutyrate, isovalerate and valerate concentrations tended ($P<0.10$) to increase, whereas acetate concentration and acetate to propionate ratio lowered ($P<0.05$) due to CPRE addition in the diet (Table 7). Molar proportion of acetate lowered ($P<0.001$), whereas

Table 3 Effect of inclusion of a composite polyphenolic-rich extract (CPRE) on in vitro feed degradability and microbial biomass production

Variable	Dose of CPRE, g/kg					SEM	P value	Contrast P value	
	0	10	20	30	40			Linear	Quadratic
NGP, mL/g at 24 h	142	145	147	148	147	1.68	0.81	0.32	0.54
IVTOMD, g/kg	721 ^a	731 ^c	731 ^c	728 ^{bc}	727 ^b	1.19	0.001	0.015	0.001
IVNDFD, g/kg	476 ^a	494 ^b	497 ^b	489 ^b	488 ^b	2.46	0.002	0.013	<0.001
PF, mg/mL	5.07 ^c	5.06 ^c	4.96 ^b	4.92 ^a	4.95 ^b	0.020	<0.001	<0.001	0.001
MBP, mg/g digested OM	408 ^c	413 ^d	407 ^{bc}	402 ^a	404 ^{ab}	0.32	0.001	<0.001	0.15

NGP net gas production, IVTOMD in vitro true organic matter degradability, IVNDFD in vitro neutral detergent fiber degradability, PF partitioning factor, MBP microbial biomass production, DM dry matter, SEM pooled standard error of mean

^{a,b,c,d}Means with different superscripts in a row differ significantly ($P<0.05$)

Table 4 Effect of a composite polyphenolic-rich extract (CPRE) on in vitro gas production kinetics and methane production

Variable	Dose of CPRE, g/kg					SEM	<i>P</i> value	Contrast <i>P</i> value	
	0	10	20	30	40			Linear	Quadratic
Gas production, mL/g DM									
2 h	4.17	4.95	4.08	6.47	4.01	0.43	0.38	0.70	0.39
4 h	13.3	15.7	13.9	17.8	12.0	1.07	0.51	0.95	0.26
6 h	25.0	29.7	30.2	33.9	28.0	1.37	0.39	0.31	0.15
8 h	40.8	44.6	48.2	51.8	45.7	1.97	0.52	0.25	0.26
10 h	58.3	67.7	62.9	68.8	63.3	1.82	0.43	0.41	0.24
12 h	77.5	83.3	84.2	87.4	81.7	1.48	0.37	0.26	0.13
24 h	140.8	141.9	143.0	142.4	140.2	0.54	0.51	0.85	0.10
36 h	169.2	171.6	162.6	173.9	157.8	1.92	0.12	0.17	0.32
48 h	175.0	177.4	177.3	181.2	170.7	1.90	0.54	0.73	0.20
60 h	179.2 ^a	188.1 ^{ab}	191.1 ^b	189.3 ^{ab}	183.5 ^{ab}	1.04	0.026	0.21	0.003
72 h	182.5 ^a	192.2 ^b	198.5 ^b	199.0 ^b	190.7 ^{ab}	0.96	0.002	0.070	<0.001
Gas production kinetics									
<i>L</i> , h	1.69	1.43	1.39	1.21	1.46	0.135	0.24	0.14	0.11
<i>Y</i> _{max} , mL/g DM	196 ^a	203 ^{ab}	207 ^b	208 ^b	199 ^a	2.25	0.015	0.20	0.002
<i>K</i> , per h	0.047	0.047	0.044	0.047	0.046	0.002	0.87	0.69	0.70
<i>T</i> _{1/2} , h	14.7	14.7	15.7	14.9	15.2	0.78	0.88	0.60	0.66
Methane production at <i>T</i> _{1/2}									
CH ₄ , mL/g DM	54.1 ^c	51.4 ^{bc}	45.6 ^{ab}	42.6 ^a	42.9 ^a	0.89	0.007	0.001	0.26
CH ₄ , mL/g IVTDDM	75.7 ^c	69.8 ^{bc}	61.3 ^{ab}	56.3 ^a	56.7 ^a	1.25	0.002	<0.001	0.16
CH ₄ , mL/g IVTDOM	80.6 ^c	74.1 ^{bc}	64.4 ^{ab}	58.8 ^a	58.9 ^a	1.46	0.003	<0.001	0.19

L lag period, *Y*_{max} maximum gas production, *K* rate of gas production, *T*_{1/2} time at which half of the maximum gas production occurs, *DM* dry matter, *CH*₄ methane, *IVTDDM* in vitro true degraded dry matter, *IVTDOM* in vitro true degraded organic matter, *SEM* pooled standard error of mean

^{a,b,c}Means with different superscripts in a row differ significantly ($P < 0.05$)

molar proportions of propionate ($P < 0.001$) and butyrate ($P < 0.001$) increased, but molar proportions of isobutyrate, isovalerate and valerate ($P > 0.10$) were not affected due to feeding of CPRE. Dietary addition of CPRE increased ($P < 0.001$) fermentation efficiency and reduced ($P < 0.001$) SCFA utilisation index. Concentrations of total N, TCA-N and non-protein N were greater ($P < 0.01$), but ammonia-N concentration decreased ($P = 0.027$) in the CPRE-fed group compared to the control group.

Relative abundances of total bacteria, total protozoa, methanogens and *R. flavefaciens* in the rumen were similar ($P > 0.10$) between two groups, but *F. succinogenes* in the rumen of buffaloes increased ($P < 0.01$) due to addition of CPRE (Fig. 1).

Plasma metabolites, enzymes and immune response (in vivo experiment)

Concentrations of different constituents in plasma, namely total protein, albumin, globulin, triglyceride, glucose, blood urea and creatinine, and enzyme activities, i.e. aspartate aminotransferase and superoxide dismutase were not altered ($P > 0.10$) by CPRE feeding (Table 8). Only concentration of cholesterol decreased

($P = 0.017$) and alanine aminotransferase activity tended to increase ($P = 0.096$) in the CPRE group compared with the control group.

Cell-mediated immunity was improved due to CPRE feeding, which was indicated by greater ($P < 0.05$) skin thickness at 12, 24 and 48 h of antigen injection, but skin thickness at 72 h was similar ($P = 0.19$) between two groups (Fig. 2).

Plasma amino acids (in vivo experiment)

Concentrations of a number of amino acids in plasma were changed (Table 9). Out of 17 amino acids analysed, for example, plasma concentrations of histidine, serine, aspartic acid, glutamic acid, threonine and cysteine were greater ($P < 0.05$) and concentrations of glycine ($P = 0.078$) and methionine ($P = 0.055$) in the CPRE-buffaloes than in the control-buffaloes. As a result, concentrations of total, total essential, total non-essential and total glucogenic amino acids increased in the plasma of buffaloes fed CPRE compared with the control diet-buffaloes. Plasma-free amino acid composition relative to plasma protein content also revealed that proportion of histidine, serine, aspartic acid,

Table 5 Effect of a composite polyphenolic-rich extract (CPRE) on in vitro ruminal concentrations and proportions of short-chain fatty acid (SCFA) concentration and concentration of ammonia

Variable	Dose of CPRE, g/kg					SEM	P value	Contrast P value	
	Control	10	20	30	40			Linear	Quadratic
SCFA concentration, mM									
Total SCFA	70.7 ^a	75.6 ^b	78.8 ^{bc}	79.4 ^{bc}	80.4 ^c	1.21	0.003	<0.001	0.024
Acetate	45.8 ^a	47.8 ^{ab}	48.5 ^{ab}	49.0 ^b	49.5 ^b	0.47	0.027	0.004	0.17
Propionate	13.4 ^a	15.7 ^b	17.4 ^b	17.2 ^b	17.8 ^b	0.56	0.003	<0.001	0.019
Butyrate	6.55 ^a	7.00 ^b	7.90 ^c	8.00 ^c	8.00 ^c	0.201	<0.001	<0.001	<0.001
BCFA	4.95	5.15	5.05	5.05	5.06	0.040	0.754	0.77	0.47
A:P	3.43 ^b	3.05 ^{ab}	2.79 ^a	2.85 ^a	2.77 ^a	0.086	0.011	0.002	0.031
Molar proportion of SCFA, %									
Acetate	64.8 ^b	63.2 ^{ab}	61.6 ^a	61.8 ^a	61.6 ^a	0.44	0.008	0.002	0.031
Propionate	18.9 ^a	20.7 ^{ab}	22.0 ^b	21.7 ^b	22.2 ^b	0.43	0.011	0.002	0.038
Butyrate	9.30 ^a	9.25 ^a	10.0 ^b	10.2 ^b	10.1 ^a	0.130	0.006	0.001	0.13
BCFA	6.95 ^b	6.80 ^b	6.40 ^a	6.40 ^a	6.25 ^a	0.092	0.004	<0.001	0.19
Fermentation efficiency									
FE, %	73.8 ^a	74.6 ^{ab}	75.2 ^b	75.1 ^b	75.3 ^b	0.20	0.007	0.001	0.027
SCFA-UI	4.29 ^b	3.85 ^{ab}	3.62 ^a	3.69 ^a	3.59 ^a	0.092	0.022	0.005	0.050
Ammonia-N, mg/L	196 ^c	194 ^c	181 ^b	178 ^a	177 ^a	0.61	<0.001	<0.001	0.006

SCFA short-chain fatty acids, BCFA total branched-chain fatty acids (isovalerate and isobutyrate), A:P acetate to propionate ratio, FE fermentation efficiency, SCFA-UI short-chain fatty acid utilisation index, SEM pooled standard error of mean

^{a,b,c}Means with different superscripts in a row differ significantly ($P < 0.05$)

glutamic acid, threonine, cysteine, total, total essential, total non-essential and total glucogenic amino acids were greater ($P < 0.05$), and methionine proportion ($P = 0.053$) tended to be greater in the buffaloes fed CPRE than in the control buffaloes (Table S1).

Estimated methane emission and environmental footprint (in vivo experiment)

Methane production expressed in terms of per unit of SCFA, DM intake, digestible OM intake, digestible NDF intake and percent of gross energy intake were lowered ($P < 0.01$) and daily methane emission tended to decrease ($P = 0.067$) in buffaloes by CPRE feeding (Table 10). Different environmental indicators, namely methane production ($P = 0.012$), excretion of faecal nitrogen ($P = 0.025$), urinary nitrogen ($P = 0.066$), total nitrogen ($P = 0.031$), faecal phosphorus ($P = 0.016$), total phosphorus ($P = 0.016$) per unit of body weight gain, were lower or tended to be lower in buffaloes fed the CPRE-diet than in buffaloes fed the control diet.

Discussion

The CPRE used in the present study comprised of dry extracts from bark of *Acacia catechu*, bark of *Acacia arabica*, peel of *Punica granatum* and seeds of *Eugenia*

jambolana at an equal ratio, which contained high amounts of several phytochemicals, i.e. phenolics, flavonoids, condensed tannins, saponins and antioxidants. These plant materials are rich in many bioactive compounds with nutraceutical properties. The extract from bark of *Acacia arabica* contains many pharmacologically active phytochemicals such as phenolic (up to 83 mg/100 g) and flavonoid (44 mg/100 g) compounds, namely quercetin, kaempferol, chlorogenic acid, ellagic acid, catechin, epicatechin, rutin, umbelliferone with anti-inflammatory, immunomodulatory, antioxidant and antimicrobial properties (Kumar et al. 2019). The bark of *Acacia arabica* is reported to contain 12 to 20% tannins along with flavonoids, phenolic acids, saponins, alkaloids and terpenoids, particularly catechin, epicatechin, dicatechin, catechin gallate epigallocatechin gallate, quercetin, isoquercetin, leucocyanadin, leucocyanidin gallate, kaempferol, rutin, gallic acid, ellagic acid and chlorogenic acid (Mohammad et al. 2014). The bark extract of this plant showed antibacterial, antifungal, antioxidant and hypoglycaemic properties (Mohammad et al. 2014; Rajvaidhya et al. 2012). The peel of *Punica granatum* contains several polyphenolic compounds including flavonoids (anthocyanins, e.g. delphinidin, cyanidin, pelargonidin and anthoxanthins, e.g. quercetin, epicatechin, catechin and their derivatives), tannins (ellagitannins, ellagic acid and their derivatives, e.g. punicalin, pedunculagin and punicalagin) and phenolic acids (ferulic, ellagic, gallic, cinnamic, chlorogenic, caffeic, syringic, sinapic and p-coumaric acid) with many biological and pharmacological activities (Singh et al. 2018). The seeds of

Table 6 Effect of inclusion of a composite polyphenolic-rich extract (CPRE; at 20 g/kg diet) to a total mixed ration on feed intake, nutrient utilisation and growth performance in growing male buffaloes

Variable	Control	CPRE	SEM	<i>P</i> value
DM intake, kg/day	4.66	4.92	0.085	0.40
GE intake, MJ/day	94.2	98.8	1.70	0.18
Body weight				
Initial, kg	155	155	2.85	0.99
Final, kg	199	205	3.16	0.39
Total weight gain, kg	44.4	50.0	1.36	0.035
Weight gain, g/d	493	555	15.1	0.035
Feed efficiency, kg gain/kg feed	0.106	0.114	0.003	0.26
Digestibility of nutrients, g/kg				
Dry matter	614	632	4.43	0.039
Organic matter	631	649	4.75	0.056
Crude protein	624	654	6.82	0.021
Neutral detergent fiber	555	569	4.88	0.16
Acid detergent fiber	474	498	7.29	0.20
Cellulose	622	641	5.55	0.070
Nitrogen utilisation, g/day				
Intake	106	109	2.27	0.51
Faecal nitrogen	39.8	37.7	1.05	0.20
Urinary nitrogen	36.1	34.4	1.36	0.27
Absorption	65.9	71.3	1.80	0.25
Retention	29.8	36.1	1.41	0.011
ABV, %	45.2	51.6	1.64	0.032
Phosphorus excretion, g/day				
Intake	17.8	18.3	0.38	0.53
Faecal phosphorus	12.8	12.1	0.25	0.17
Urinary phosphorus	0.15	0.16	0.008	0.45
Absorption	5.05	6.26	0.27	0.020
Retention	4.90	6.10	0.27	0.020
Apparent absorption, %	28.3	34.0	1.06	0.003

DM dry matter, GE gross energy, ABV apparent biological value, SEM pooled standard error of mean

Eugenia jambolana contain various bioactive compounds including flavonoids, terpenes and anthocyanins with a range of pharmacological activities, e.g. antimicrobial, anticancer, hepatoprotective, anti-inflammatory, free radical scavenging and antidiabetic effects (Baliga et al. 2011). These plant materials are integral parts in various ethnomedicines. Due to rich concentrations of several types of bioactive compounds with especially antimicrobial, antioxidant and immune-modulatory actions, this CPRE exerted beneficial responses on ruminal fermentation, growth performance and immunity in buffaloes.

In the in vitro study, net gas production was similar among treatments up to 24 h, but was greater at 60 or 72 h and feed degradability increased due to addition of CPRE usually at doses of 20 to 30 g/kg substrate. This indicates

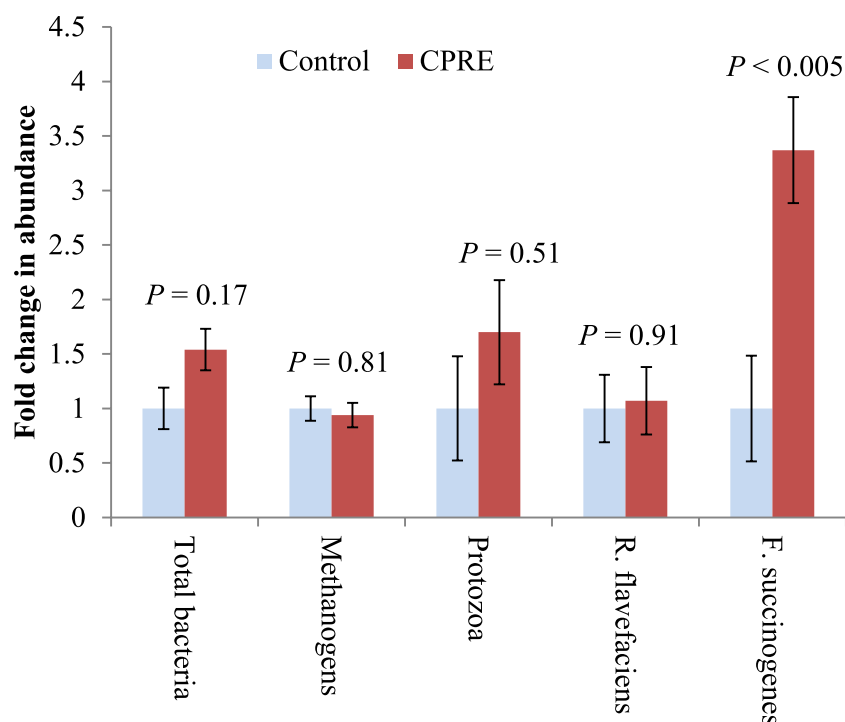
Table 7 Effect of a composite polyphenolic-rich extract (CPRE; at 20 g/kg diet) on concentrations and proportion of short-chain fatty acid (SCFA) and nitrogen fractions in ruminal fluid of growing male buffaloes

Variable	Control	CPRE	SEM	<i>P</i> value
SCFA concentration, mM				
Total SCFA	101	106	1.01	0.026
Acetate	63.5	60.5	0.71	0.027
Propionate	19.1	22.3	0.41	<0.001
Isobutyrate	3.19	3.36	0.165	0.056
Butyrate	11.1	14.4	0.51	<0.001
Isovalerate	2.83	3.26	0.114	0.059
Valerate	1.36	1.75	0.110	0.074
A:P	3.33	2.72	0.065	<0.001
Molar proportion of SCFA, %				
Acetate	62.8	57.2	0.59	<0.001
Propionate	18.9	21.1	0.29	<0.001
Isobutyrate	3.16	3.20	0.153	0.92
Butyrate	10.9	13.7	0.42	<0.001
Isovalerate	2.81	3.10	0.112	0.19
Valerate	1.34	1.65	0.104	0.19
Fermentation efficiency, %				
FE	74.1	75.7	0.16	<0.001
SCFA-UI	4.27	3.80	0.069	<0.001
Nitrogen fractions, mg/L				
Total nitrogen	787	869	8.28	<0.001
TCA-N	514	567	6.89	0.001
NPN	273	302	3.98	<0.001
NH ₃ -N	261	248	3.09	0.027

SCFA short-chain fatty acids, A:P acetate to propionate ratio, FE fermentation efficiency, SCFA-UI short-chain fatty acid utilisation index, SEM pooled standard error or mean, TCA-N trichloroacetic acid precipitable nitrogen, NPN non-protein nitrogen, NH₃-N ammonia nitrogen, SEM pooled standard error of mean

that this additive mixture stimulated the microbial activities leading to high feed degradability. Greater DM digestibility was also noted in the present in vivo study on buffaloes, which resulted from increased CP digestibility and also to some extent cellulose digestibility. Dietary phytochemicals modulate ruminal fermentation and microbiome structures depending upon the dose and nature of compounds and diet type (Patra and Saxena 2009). In the present study, the additive mixture contained phenolic compounds that had a positive impact on the ruminal microbiota and fermentation at doses of 20 to 30 g/kg diet. The stimulation of the anaerobic microbial population may result from free radical scavenging activities and probiotic effects of the phenolic compounds in the CPRE. Various polyphenols including the extracts used in the present additive act as strong free radical scavengers (Rice-Evans 1995). Rumen anaerobic microbiota sometimes faces oxidative stress, especially immediately after feed and

Fig. 1 Effect of feeding a composite polyphenol-rich extract (CPRE) on microbial population changes (fold changes compared with the control group) in male buffalo rumen



water intake and during this time anaerobic microbial activities may be reduced (Scott et al. 1983). However, polyphenols with strong ant-oxidative actions may ameliorate the oxidative stress quickly, leading to stimulation of the strictly anaerobic microbiota. Red chicory polyphenol compounds with antioxidant activities appear to stimulate ruminal microbial growth in the in vitro conditions (Cattani et al. 2012). Polyphenolic compounds and their metabolites, probably as prebiotics, also stimulate growth of specific groups of microorganisms such as *Bifidobacteria*, *Lactobacillus*

Table 8 Effect of inclusion of a composite polyphenolic-rich extract (CPRE; at 20 g/kg diet) to a total mixed ration on the blood biochemical indices of growing male buffaloes

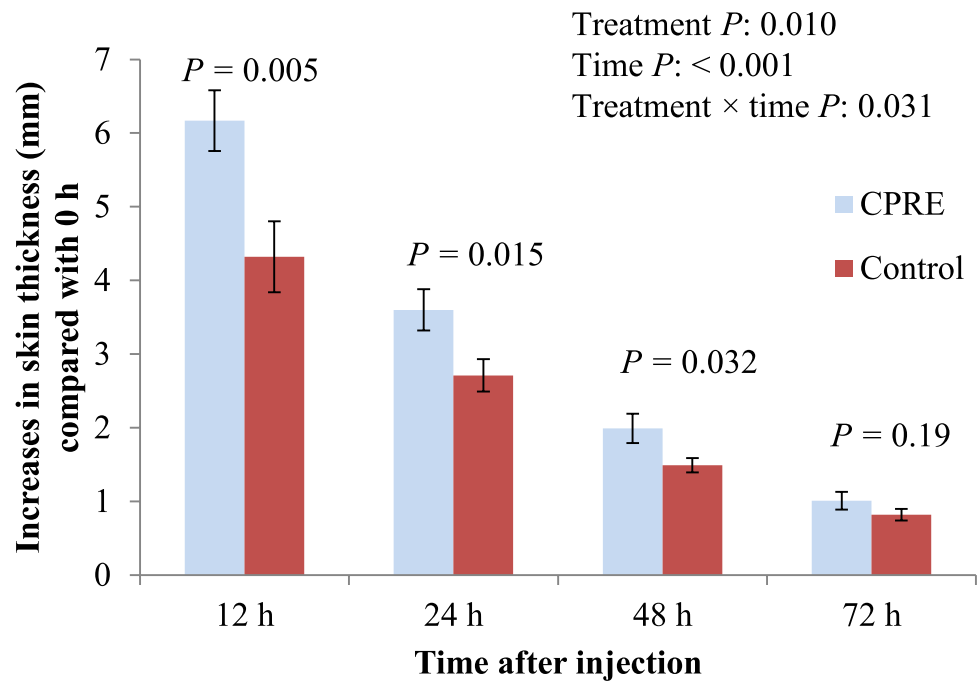
Variable	Control	CPRE	SEM	P value
Glucose, g/L	526	571	23.6	0.36
Total protein, g/L	74.0	75.6	1.57	0.62
Albumin (A), g/L	26.6	29.1	0.79	0.12
Globulin (G), g/L	47.4	46.5	1.29	0.75
A:G	0.57	0.63	0.03	0.24
Cholesterol, g/L	632	523	23.8	0.017
Triglycerides, g/L	293	306	16.5	0.70
BUN, g/L	259	238	8.53	0.24
Creatinine, g/L	3.35	3.55	0.11	0.36
AST, IU/mL	96.7	96.4	2.74	0.97
ALT, IU/mL	34.3	38.3	1.22	0.096
SOD, IU/mL	10.5	10.1	0.27	0.43

BUN blood urea nitrogen, AST aspartate aminotransferase, ALT alanine aminotransferase, SOD superoxide dismutase, SEM pooled standard error of mean

and *Akkermansia* in the gut (Zhou et al. 2020). In pure culture study with ruminal bacteria, some phenolic acids stimulated the growth rate of bacteria in species- and dose-specific manners—syringic acid, *p*-hydroxybenzoic acid and hydroxycinnamic acid increased growth rate of *R. albus*, *R. flavefaciens* and *Butyrivibrio fibrisolvens* (Borneman et al. 1986). In the present study, numbers of total bacteria, total methanogens, total protozoa and *R. flavefaciens* were not affected, but cellulolytic *F. succinogenes* population increased by CPRE. High concentrations of polyphenolic compounds including tannins, flavonoids and phenolic acids are always toxic ruminal bacteria (Patra and Saxena 2011; Borneman et al. 1986). Therefore, doses of CPRE > 30 g/kg diet did not exhibit further increases in substrate degradability, potential gas production and microbial biomass synthesis, which were similar to the control in the current study. Nonetheless, these variables did not decrease beyond the control values because of use of the CPRE at doses that are not likely toxic to overall ruminal microbial degradation activities due to high redundancy of the metabolic functions (Patra 2020), but a few select microbial populations may be sensitive to CPRE at the high doses (Borneman et al. 1986).

In the present study, increased in vitro substrate DM degradation or in vivo DM digestibility was associated with increased concentrations of total SCFA both in vitro and in vivo experiments. Increased SCFA concentrations resulted from increases of all SCFA in vitro and all SCFA in vitro except acetate. However, proportion of propionate and butyrate elevated and acetate proportion decreased. This result signifies that this CPRE improved all types of

Fig. 2 Effect of inclusion of a composite polyphenol-rich extract (CPRE) to the total mixed ration on the cell mediated immune response (increases in skin thickness, mm) in growing male buffaloes



fermentative activities of the microbiota, although there were preferential influences on the microbial communities particularly on the propionate and butyrate producers. This is further substantiated from the stimulation of *F. succinogenes*, an acetate and succinate (precursor of propionate) producer, but *R. flavefaciens*, a acetate and butyrate producer (Yeoman et al. 2021), was not affected in the present study. The increase in molar proportion of propionate that is energetically more efficient than acetate and butyrate is responsible for improved fermentation efficiency and this is expected to increase animal productivity.

Besides improving fermentation efficiency, this CPRE decreased the ammonia concentration in the rumen in both in vitro test and in buffaloes. Excessive ammonia production in the rumen arises due to degradation of dietary protein, which is absorbed from the rumen and excreted through urine as urea (Dijkstra et al. 2013; Patra 2015). This represents dietary protein loss, body energy loss as well as environmental burdens as nitrogen excretion and nitrous oxide emissions (Dijkstra et al. 2013). Therefore, many strategies, namely protein treatment, dietary use of ruminal undegradable protein and ruminal microbial modulation, have been proposed to reduce excessive degradation of dietary protein (Dijkstra et al. 2013). Lowered ammonia concentration in the rumen results from formation of protein–phenolic compound complex, inhibition of predominant protein degrading or ammonia producing microbiota and use of ammonia for synthesis of microbial protein (Lagrange and Villalba 2019; Patra and Saxena 2011). In the present study, decreased ammonia concentration is unlikely contributed from decreased deamination from amino acids, because

concentrations of branched-chain fatty acids (i.e. isobutyric, isovaleric, and 2-methylbutyric acid) that are mainly formed from deamination of branched-chain amino acids by ruminal microbes (Apajalahti et al. 2019) were not significantly altered. Therefore, phenolic compounds along with tannins perhaps formed complexes with dietary proteins, leading to reduced degradation in the rumen, which was available post-ruminally (Lagrange and Villalba 2019; Patra et al. 2012) because protein digestibility did not lower rather it increased in vivo. Reduced ammonia concentration may also be caused by increased microbial protein synthesis using amino acids or ammonia for their growth. Increased estimated microbial biomass was noted in the present in vitro experiment, which was further substantiated from greater concentrations of ruminal TCA-precipitable nitrogen (which represents true protein from microbial protein and dietary protein) in the in vivo experiment. Although ammonia concentration in the rumen reduced in the CPRE group, nonprotein nitrogen concentrations, which possibly represented amino acids and smaller peptides, were higher in this group compared with the control, which may arise from higher amino acids and peptides pool from microbial origins.

Further, this CPRE mitigated methane production in vitro and in buffaloes, which was probably attributed to altered ruminal fermentation, i.e. predominantly increased propionate production. Propionate acts as a hydrogen sink redirecting hydrogen from methanogenesis to propionigenesis. Polyphenols, flavonoids, tannins and saponins decrease methane production directly inhibiting the methanogens, protozoa-related methanogens or lowering hydrogen production from fiber degradation (Patra and Saxena 2010; Patra et al. 2017),

Table 9 Effect of inclusion of a composite polyphenolic-rich extract (CPRE; at 20 g/kg diet) to the total mixed ration on the plasma free amino acid concentrations (μM) in growing male buffaloes

Variable	Control	CPRE	SEM	P-value
Individual amino acids				
Histidine	274	357	13.8	0.001
Serine	56.2	91.9	65.9	<0.001
Arginine	39.3	39.0	0.28	0.58
Glycine	255	272	4.80	0.078
Aspartic acid	11.6	16.8	0.67	<0.001
Glutamic acid	93.4	118	4.52	0.003
Threonine	32.5	45.2	3.05	0.033
Alanine	118	128	3.96	0.22
Proline	35.1	38.1	1.29	0.25
Cysteine	13.5	18.4	0.90	0.003
Lysine	43.3	41.4	1.83	0.61
Tyrosine	29.7	34.1	2.07	0.30
Methionine	64.7	76.3	3.06	0.055
Valine	141	138	5.69	0.82
Isoleucine	58.5	59.3	2.18	0.86
Leucine	44.7	44.2	2.18	0.90
Phenylalanine	50.0	48.8	2.44	0.82
Group of amino acids				
ΣEAA ^a	747	849	18.5	0.003
ΣNEAA ^b	613	718	15.9	<0.001
ΣTAA ^c	1360	1567	30.8	<0.001
ΣKAA ^d	88.1	85.6	2.81	0.67
ΣGAA ^e	1134	1339	30.2	<0.001
ΣBCAA ^f	244	242	6.75	0.86

EAA essential amino acid, NEAA non-essential amino acid, TAA total amino acid, KAA ketogenic amino acid, GAA glucogenic amino acid, BCAA branched-chain amino acid, SEM pooled standard error of mean

^aEAA: Thr + Val + Met + Ile + Leu + Phe + His + Lys + Arg

^bNEAA: Ser + Glu + Asp + Pro + Gly + Ala + Cys-Cys + Tyr

^cTAA: EAA + NEAA

^dKAA: Leu + Lys

^eGAA: Thr + Val + Met + His + Arg + Ser + Glu + Gly + Ala + Pro + Asp + Cys-Cys

^fBCAA: Val + Ile + Leu

but a decrease in methane production was unlikely due to these reasons in the present study because methanogenic *mcrA* gene expression, total protozoal abundance and fiber degradation were not altered in the present in vivo experiment. Polyphenolic extracts or feeds from different plant sources also reduced methane production (Huang et al. 2021; Sinz et al. 2019). The extract of *Punica granatum* at 20 g/kg diet for buffaloes lowered methane production by 25.3% (Hundal et al. 2019). In the in vitro study, *Acacia catechu* extract decreased methane production (Wadhwa et al. 2020). Dietary supplementation of *Syzygium cumini* fruit shreds at

Table 10 Effect of inclusion of a composite polyphenolic-rich extract (CPRE; at 20 g/kg diet) to a total mixed ration on estimated methane emissions and environmental footprint in growing male buffaloes

Variable	Control	CPRE	SEM	P value
Methane production				
g/mol SCFA ^a	4.39	4.07	0.031	<0.001
g/kg DM intake ^b	20.5	18.7	0.18	<0.001
g/kg digestible OM intake	32.6	28.8	0.37	<0.001
g/kg digestible NDF intake	37.0	32.8	0.44	<0.001
MJ/d	5.37	5.02	0.125	0.067
% of GE	5.44	5.08	0.074	0.003
g/kg ADG	200	163	9.39	0.012
Nitrogen excretion				
Faecal nitrogen, g/kg ADG	80.8	67.9	3.709	0.025
Urinary nitrogen, g/kg ADG	73.4	62.1	4.251	0.066
Total nitrogen, g/kg ADG	154	130	7.437	0.031
Phosphorus excretion				
Faecal phosphorus, g/kg ADG	26.1	21.8	1.139	0.016
Urinary phosphorus, g/kg ADG	0.31	0.29	0.019	0.387
Total phosphorus, g/kg ADG	26.4	22.1	1.153	0.016

DM dry matter, GE gross energy, SCFA short-chain fatty acids, OM organic matter, NDF neutral detergent fiber, ADG average daily body weight gain, SEM pooled standard error of mean

^aMethane production was estimated based on the stoichiometric relationship of in vivo SCFA production = $0.50 \times \text{acetate} - 0.25 \times \text{propionate} + 0.50 \times \text{butyrate}$ (Moss et al. 2000)

^bMethane production (g/kg DM intake) was estimated based on equation from in vivo ruminal short-chain fatty acid profile = $3.28 \times (\text{acetate} + \text{butyrate})/\text{propionate} + 7.6$ (Williams et al. 2019)

20 g/kg diet lowered methane production by 29% in cross-bred lactating cattle (Mayuresh et al. 2019). In CPRE-fed buffaloes, estimated methane emission per mole of SCFA production was reduced by 7.29%. We confirmed from the in vitro experiment that estimated methane production was highly correlated ($R^2 = 0.84$) with actual methane production. Methane production per unit of fiber digestibility was also lowered by 11.4% due to CPRE feeding, suggesting that methane inhibition was not associated with decreased fiber digestibility by ruminal microbiota.

Polyphenolic compounds including tannins sometimes decrease feed intake due to astringency of tannins and reduced fiber degradability in the rumen (Patra and Saxena 2011). In the present study, feed intake was similar between two groups suggesting no effect of CPRE on the palatability. Average daily weight gain enhanced due to feeding of CPRE, which was likely due to combined outcomes of improved ruminal fermentation, microbial protein synthesis, DM digestibility and better immune responses.

Nutrient balance in the body is an indicator of the diet quality. Feeding of CPRE improved nitrogen and phosphorus retention in buffaloes. Nitrogen intake and its excretion through faeces and urine were similar between two groups,

but apparent biological value of the absorbed protein was better in the CPRE group. This suggested that absorbed protein had better amino acid balance, probably due to greater microbial protein synthesis that usually has better protein quality compared to average dietary proteins fed to ruminants. This was reflected in the greater amino acid concentrations in plasma of buffaloes. For phosphorus balance, CPRE feeding increased phosphorus absorption percent, leading to its greater retention in the body of buffaloes. To our knowledge, the effect of phenolic-rich additives on phosphorus absorption in ruminants is scarce. The reason for increased phosphorus absorption by CPRE feeding is unclear, but polyphenolic compounds may improve gut health, barrier function and nutrient absorption including phosphorus from the intestine of livestock (Patra 2018; Patra et al. 2019).

Polyphenolics and flavonoids in diets can boost immunity, antioxidant balance and productivity in ruminants (Oh et al. 2017; Olagaray and Bradford 2019). The effect of CPRE was investigated for apparent toxic effects and health benefits. Any blood picture such as protein, lipid, glucose and liver enzyme markers was not adversely affected. Rather cell-mediated immunity in buffaloes improved and cholesterol concentration decreased due to feeding of CPRE. This feed additive was rich in phenolic compounds including flavonoids, which have shown to boost endogenous antioxidant systems along with catalase, glutathione peroxidase and superoxide dismutase and by up-regulating genes and transcription factors, e.g. nuclear factor E2-related factor and aryl hydrocarbon receptor related to antioxidant stress (Oh et al. 2017; Seymour et al. 2013). Chestnut tannins (mostly hydrolysable tannins) lowered malondialdehyde content (an oxidative stress marker) and elevated superoxide dismutase and glutathione peroxidase activity in the blood and liver of transition cows (Liu et al. 2013). In the present study, superoxide dismutase activity in plasma was not affected by CPRE containing rich sources of phenolic compounds, which might be attributed to the degradation in the rumen and intestine, low dose use and high health status (i.e. low oxidative stress) of buffaloes. Similar to the present study, improved humoral and cell-mediated immunity was also reported due to feeding of pomegranate extract (15 and 30 mg gallic acid equivalent/kg body weight) to calves (Oliveira et al. 2010). Decreased cholesterol concentrations in plasma may be mediated through higher bile acid excretion in faeces and enhancing cholesterol metabolism pathways (Liu et al. 2015).

Polyphenol-rich extracts can improve lymphocyte functions, mitogen-induced lymphocyte cytokine releases and balancing between oxidants and antioxidants in immune cells, leading to enhanced immune response (Gracious et al. 2001; Oliveira et al. 2010). Finally, dietary CPRE reduced indicators of environmental impacts as noted from decreased

methane production (absolutely or per unit of body weight gain), nitrogen and phosphorus excretion per unit of body weight gain. Estimated methane yield (g/kg feed dry matter intake) was reduced by 8.78%, and methane emission intensity (per unit of body weight gain) by 18.5% in buffaloes fed CPRE. Environmental footprints (i.e. emissions or pollution per unit of useful product) need to be lowered to fulfil the demands of growing human populations (Patra 2017). Substantially greater reduction of methane intensity was resulted from combined effect of methane production and increased average daily body weight gain. Many feed additives have been explored to reduce methane yield and intensity in livestock production, but they sometimes exert deleterious effects on feed utilisation and production performance (Patra et al. 2017). A recent study reported that seaweeds (*Asparagopsis armata*) supplementation at 10 g/kg diet lowered methane intensity of milk production by 60%, but feed intake by 38% and milk yield by 11.6% also reduced; however, at 5 g/kg diet, it reduced methane intensity by 27% without affecting milk yield and feed intake in a short-term experiment (Roque et al. 2019). For sustainable livestock production, it is essential that productivity, carbon footprint of livestock products, health and welfare of the livestock are not compromised. The present study evidently demonstrated that partial carbon footprint of the product (i.e. per unit of weight gain) was lower in buffaloes on a predominantly forage based diet without affecting feed intake and health of animals. Moreover, this feed additive lowered nitrogen and phosphorus excretion per unit of body weight gain. There are limited studies investigating three most environmental impacts (i.e. methane, nitrogen and phosphorus) of livestock production by using natural feed additives, which are highly necessary to examine together to reduce overall environmental burdens of livestock production (Pelletier and Tyedmers 2010). The present study clearly exhibited that this type of CPRE may be useful to reduce environmental costs of ruminant production.

Conclusions

The study suggested that the CPRE from *Acacia arabica* bark, *Acacia catechu* bark, *Punica granatum* peel and *Eugenia jambolana* seeds mixed in an equal proportion and added at 20 g/kg diet improved body weight gain, ruminal fermentation, feed digestibility, immunity and plasma amino acids profile. This CPRE simultaneously reduced methane production and excretion of nitrogen and phosphorus per unit of body weight gain in buffaloes. Therefore, this composite polyphenol-rich feed additive at 20 g/kg diet may be used to improve growth performance and health benefits while reducing indicators of environmental impacts of buffalo production. This CPRE should be tested in high

producing lactating cattle or beef cattle production system with large number of animals to confirm the production performance and environmental benefits.

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Authors' contribution JSH and SS conceived and designed research. SS and JSH conducted experiments. SS, RSS and AS performed laboratory analyses. JSH and AKP analysed the data. AKP and JSH wrote the manuscript.

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Data availability Most of the data are presented in the tables. The data in the figure can be available from the corresponding author after a reasonable request.

Declarations

Ethics approval The study was conducted after getting the necessary approval from the Institutional Animal Ethics Committee of the University (IAEC/2020/200–219/56/1 dated 14.12.2020) working under “the Committee for the Purpose of Control and Supervision of Experiments on Animals”, New Delhi, India.

Consent to participate Not applicable.

Consent for publication Not applicable.

Conflict of interest The authors declare no conflict of interest.

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